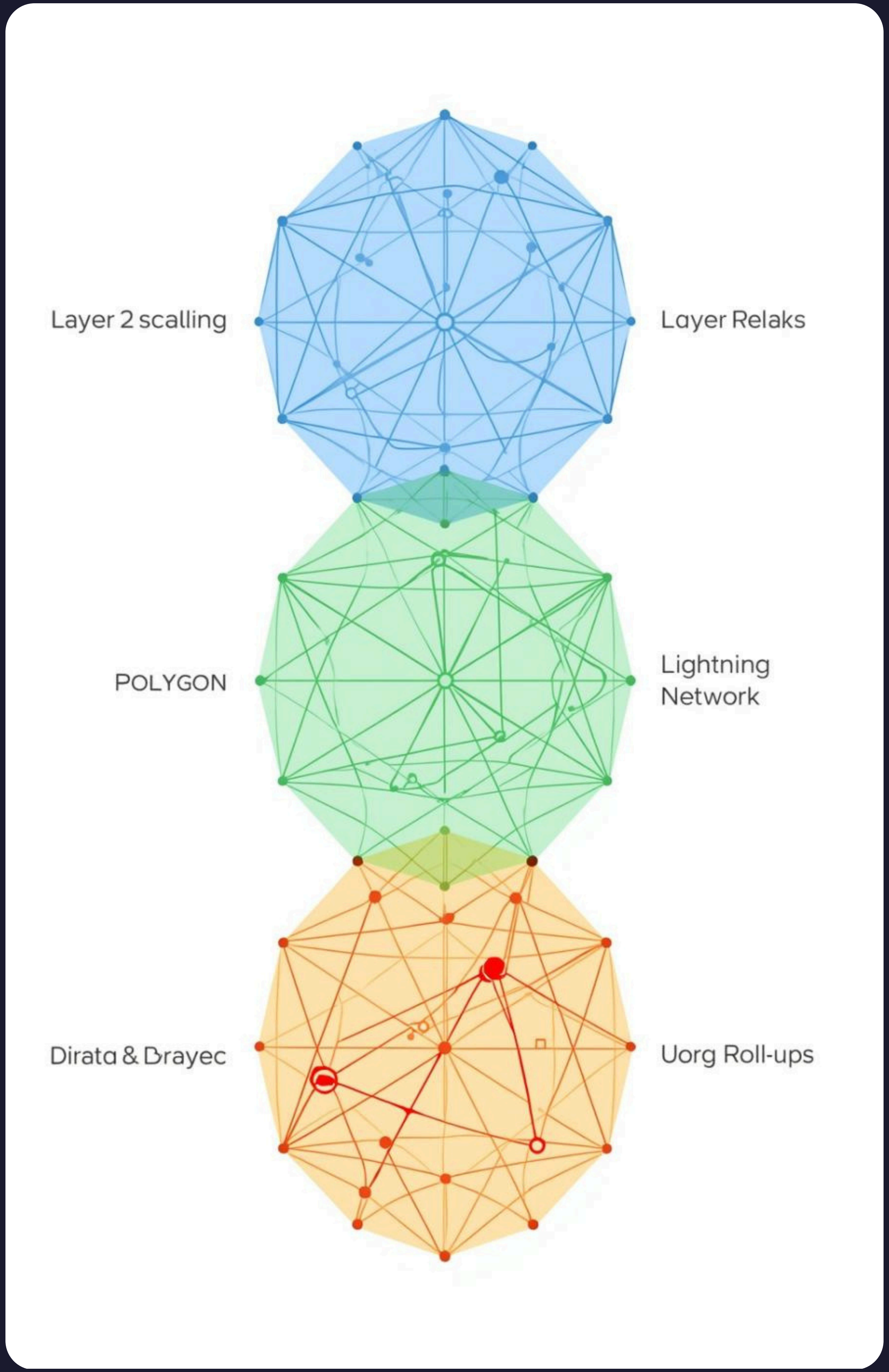
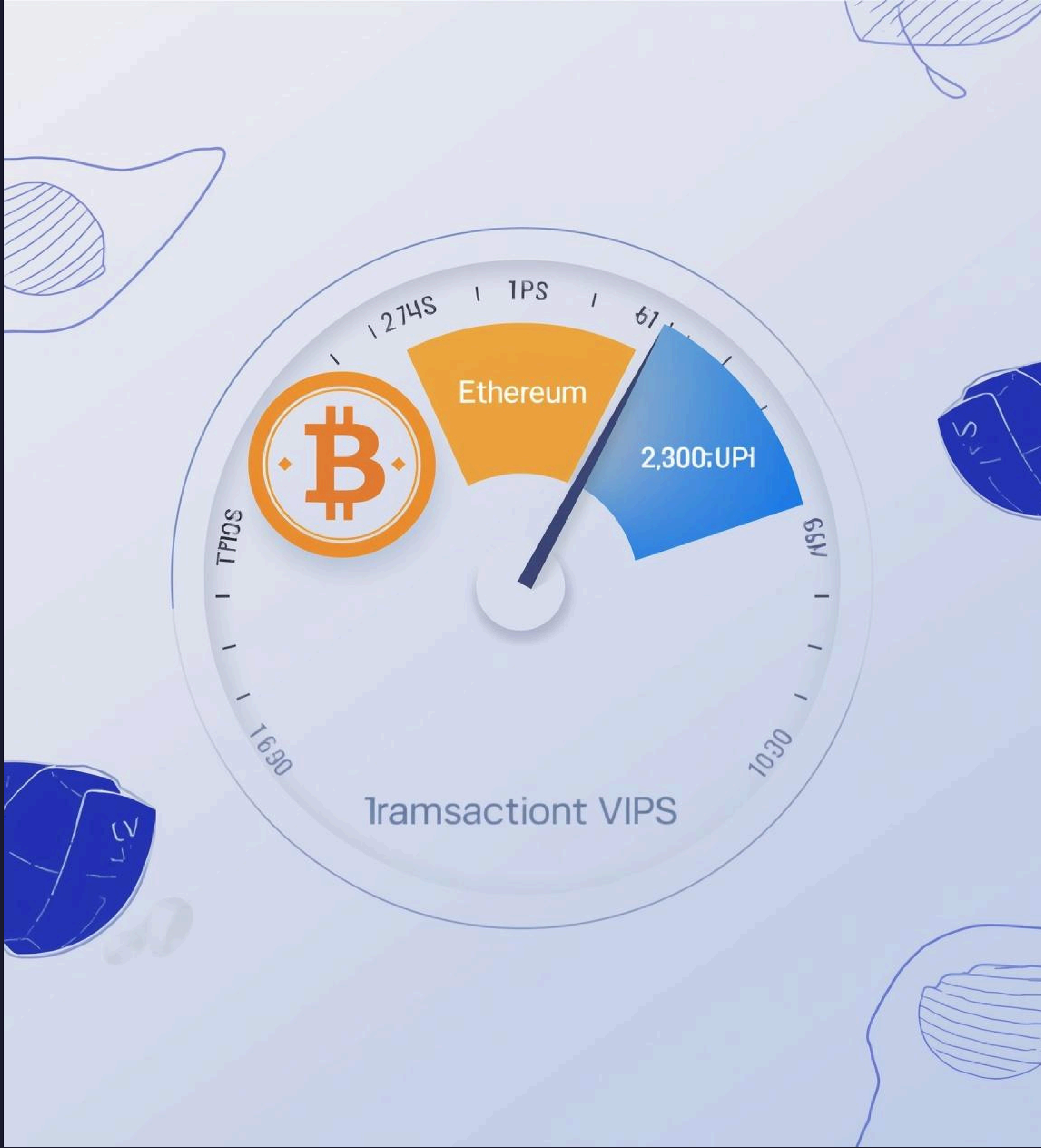
Layer-2 Scaling Solutions

Presented by Manish Raje 67



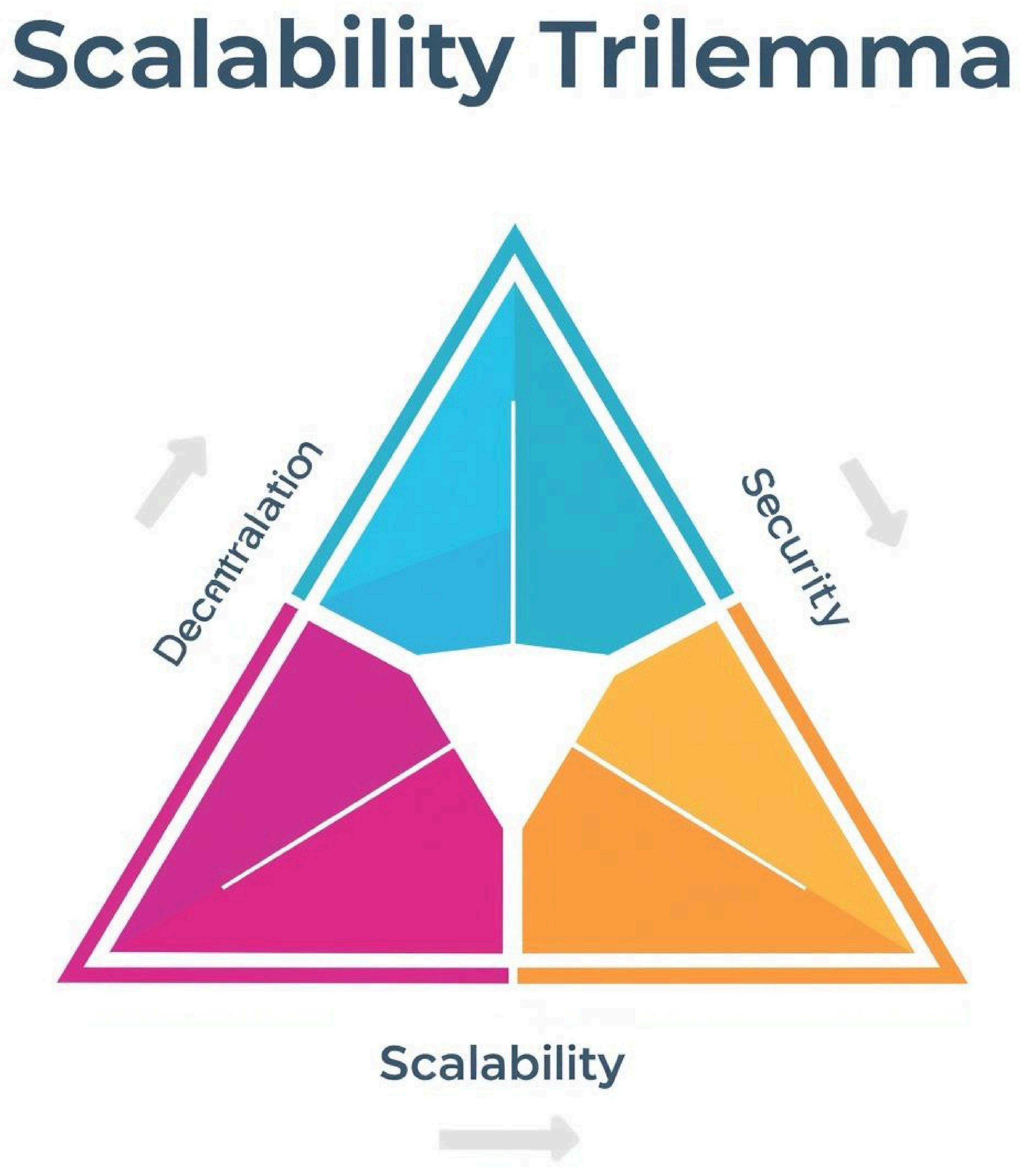
Understanding the Scalability Challenge

- Bitcoin processes ~7 transactions per second (TPS).
- Ethereum handles ~15–30 transactions per second (TPS).
- Visa manages over 2,000 transactions per second (TPS) with 65,000+ peak capacity.
- Layer-1 blockchains face a "Trilemma" balancing security, decentralization, and speed.
- High network demand results in transaction congestion and expensive gas fees.
- Layer-2 solutions are required to move high-volume processing off the main chain.
- Scalability is the primary barrier to global retail and enterprise blockchain adoption.
- Layer-2 integration enables "Visa-level" performance while maintaining network security.



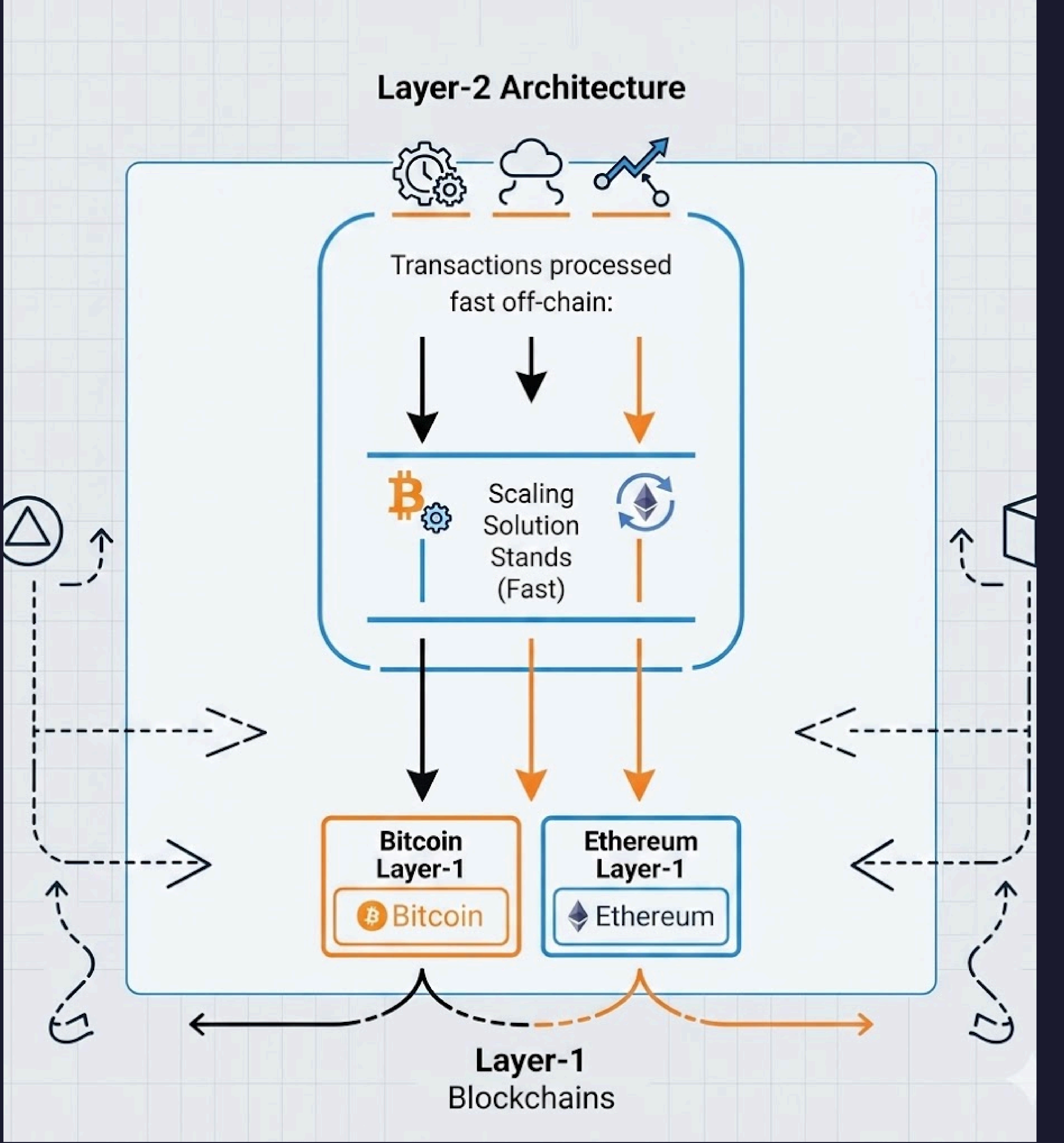
Understanding Layer-1 Limitations

- Layer-1 blockchains are limited by the physical speed of global node synchronization.
- Increased network usage leads to significant latency and high transaction costs.
- The "Blockchain Trilemma" prevents Layer-1 from scaling without losing security or decentralization.
- Layer-2 solutions provide the necessary infrastructure for high-speed, low-cost operations.
- Moving transaction execution off-chain preserves the integrity of the underlying Layer-1.
- These advancements are the key requirement for the next phase of blockchain evolution.



Understanding Layer-2 Solutions

Layer-2 solutions enhance blockchain scalability by processing transactions off-chain while maintaining security and finality through the base Layer-1 blockchain, enabling faster, cheaper, and more efficient transactions.



Polygon: A Layer-2 Aggregator

- Polygon (formerly Matic Network) functions as a high-performance sidechain for Ethereum.
- The network achieves speeds of up to 65,000 transactions per second (TPS).
- Users benefit from near-zero transaction fees compared to Ethereum Layer-1.
- Developers can migrate existing Ethereum DApps and projects without making code changes.
- It utilizes a Proof-of-Stake (PoS) consensus mechanism to secure the network.
- Polygon significantly reduces network congestion while maintaining compatibility with the Ethereum Virtual Machine (EVM).
- The platform serves as a critical scaling hub for DeFi, NFTs, and Web3 gaming.

WHAT IS POLYGON?

Polygon (previously Matic Network in 2017) is a crypto currency founded by 3 indians and has seen its market cap surge ten-fold since February, owing to increased adoption of its blockchain by players in gaming, non-fungible tokens (NFTs), and DeFi (decentralised finance). In March, Nasdaq-listed Coinbase allowed users to trade the Polygon coin.

Founders-Sandeep Nailwal(Indian), Jayanti Kanani(Indian), Anurag Arjun(Indian), Mihailo Bjelic(Serbian)

Polygon (MATIC)

Layer 2 scaling solution - scalability and usability of public blockchains (Ethereum)

faster and cheaper transactions

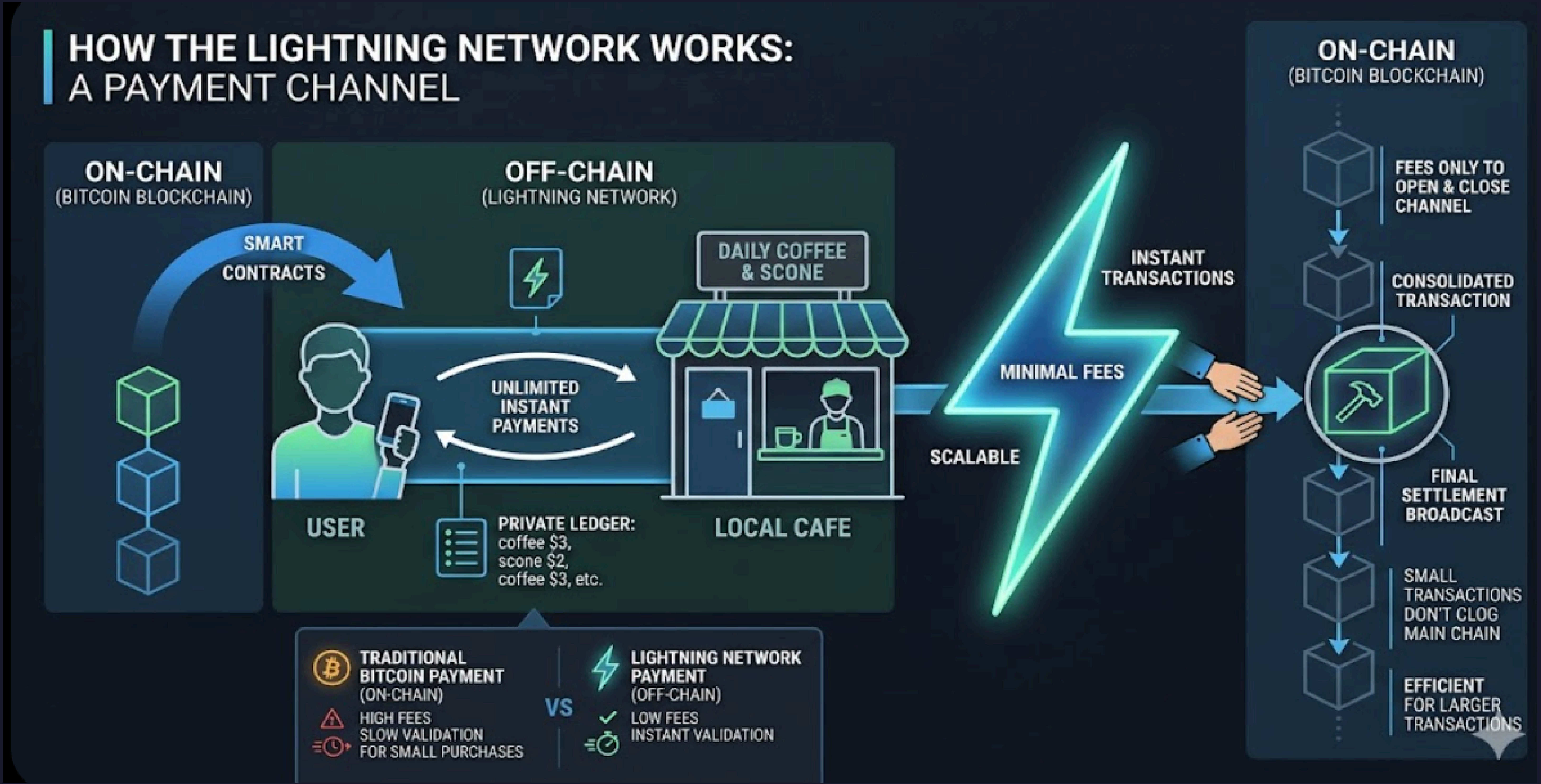
Features

- Scalability
- High throughput (10,000 TPS)
- Improved user experience
- Enhanced security
- Public, permissionless side chains

Introduction to the Lightning Network

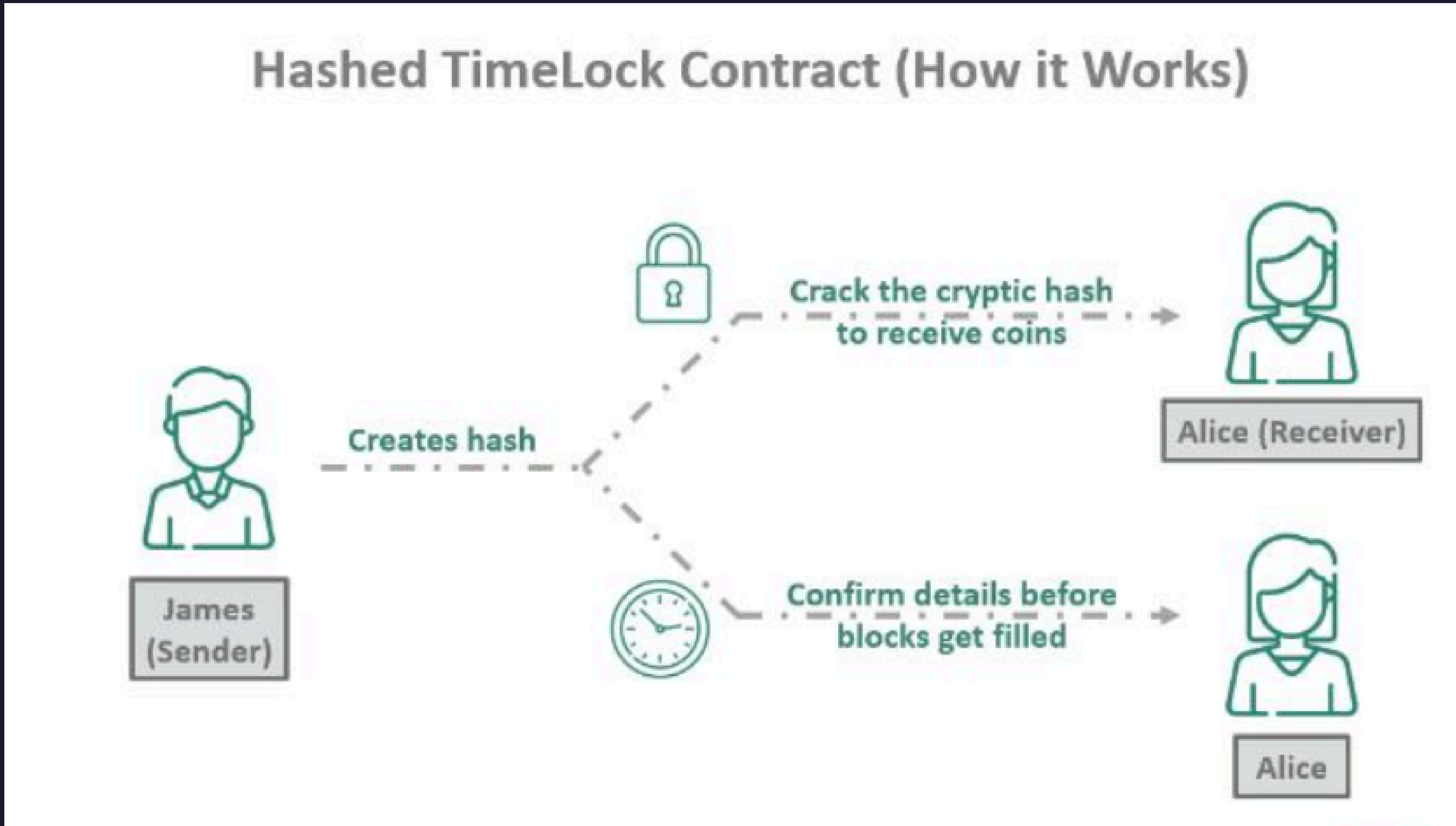
The Lightning Network is a revolutionary **Layer-2 solution** for Bitcoin, designed to enhance transaction speed and scalability, enabling instant payments while reducing congestion on the main Bitcoin blockchain.

Understanding Payment Channels in Lightning



Payment channels enable **instant, off-chain transactions** between Alice and Bob. Funds are locked in a multi-signature wallet, allowing unlimited exchanges without on-chain fees until the channel is closed, ensuring fast and efficient transactions.

Routing Payments in Lightning Network



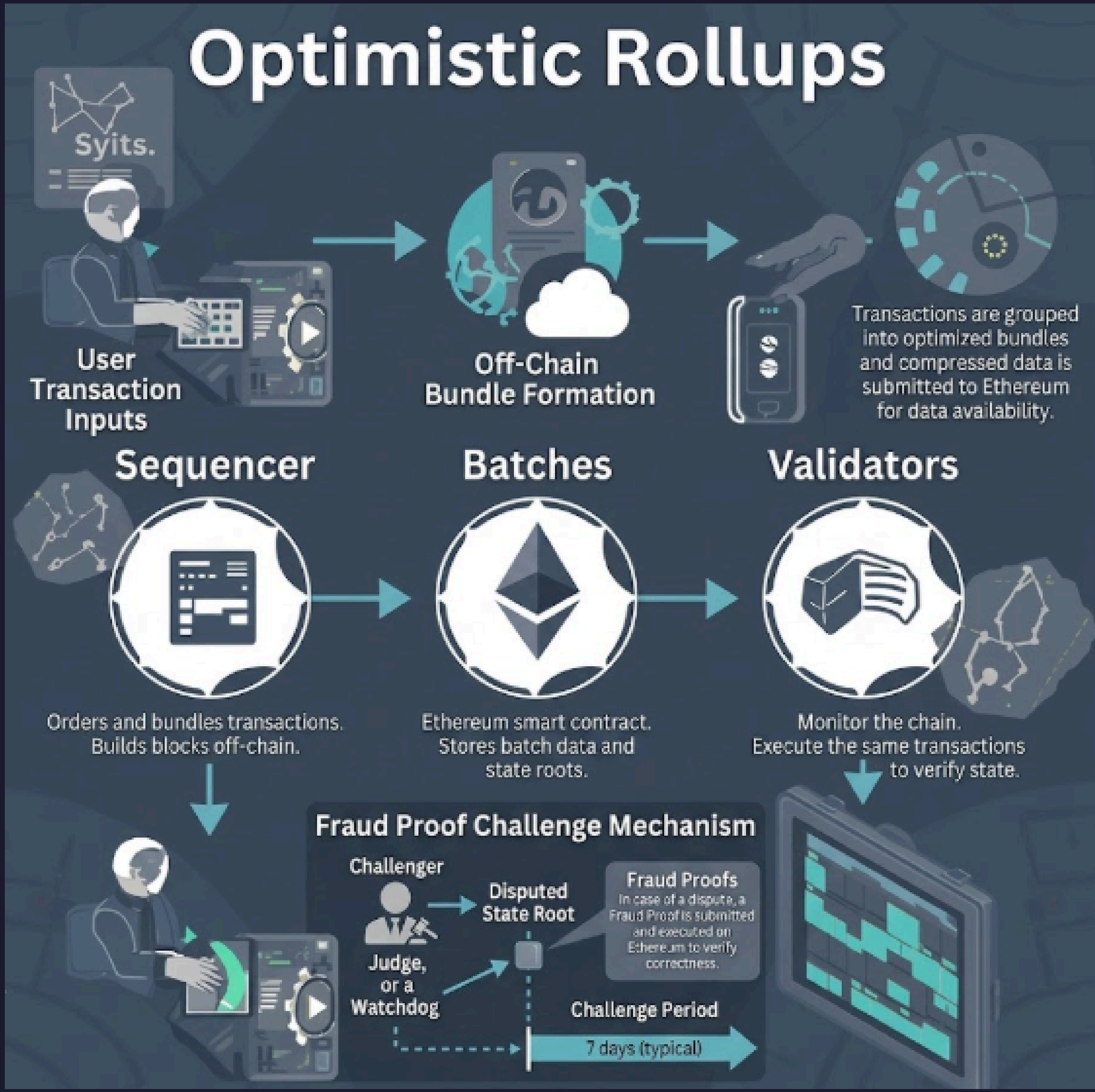
The Lightning Network efficiently routes payments using **intermediate channels**, ensuring secure and instant transactions. By leveraging **Hash Time-Locked Contracts**, it optimizes pathfinding and enables trustless payments across multiple nodes.

Understanding Optimistic Rollups

- Multiple transactions are bundled into a single batch to be processed off-chain.
- Security is maintained by using fraud proofs to challenge invalid transactions.
- The system assumes transactions are valid by default to increase processing speed.
- High-volume batching significantly reduces individual gas fees for users.
- All data is posted back to the main Ethereum chain to ensure robust security standards.
- The architecture provides a high-performance environment ideal for complex DeFi applications.
- A dispute window allows the network to roll back any malicious activity before finality.

Understanding Optimistic Rollups Architecture

Optimistic Rollups bundle transactions off-chain, submitting them to Ethereum for finality. They employ a challenge mechanism to ensure security, allowing anyone to dispute invalid transactions within a predefined timeframe.



OPTIMISTIC ROLL-UP PROJECTS:

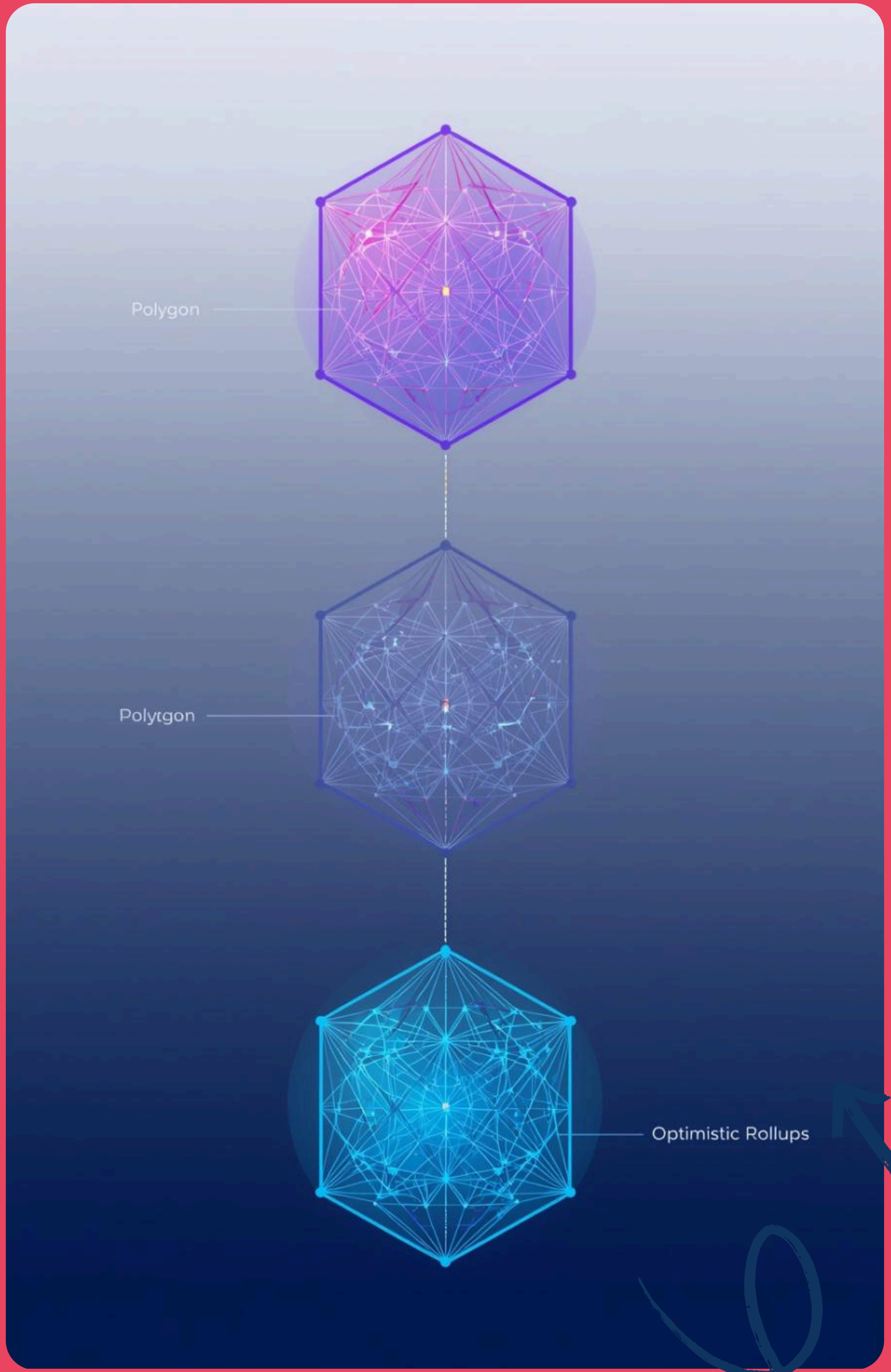
Optimistic Rollups will play a key role for the future of Ethereum.

Promising projects are:

- Arbitrum
- Optimism
- Metis Andromeda
- Fuel Network
- Boba Network

Real-World Layer-2 Adoption

Layer-2 solutions like Polygon and Lightning Network are revolutionizing industries, with **real-world applications** seen in Instagram NFTs, El Salvador's Bitcoin adoption, and platforms like Uniswap enhancing transactions and user experiences.



Conclusion

"Layer-2 is the future of scalability"

*~ **Manish Raje***